

Justin Bacic

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Education

Master of Secondary Teaching

University of Canberra | Feb 2026 – Dec 2027 | Canberra

Bachelor of Computer Science Honours Algorithms Stream

Carleton University | Sept 2021 – Apr 2025 | Ottawa

GPA 3.74, Minor in Mathematics

Skills

Languages: Python, Java, C++, C, JavaScript, SQL, HTML, Haskell, C#, CSS

Tools & Frameworks: Linux, Git, Flask, PostgreSQL, Qt, NodeJs, Power Automate, SharePoint, Docker, MatLab

Knowledge of: Data Structures, Algorithms, Database Management Systems, Web Development, Network Protocols, Operating Systems, Distributed Algorithms, Quantum Computing, REST API's Embedded Development

Work Experience

Teaching Assistant – Programming Paradigms & Algorithms

Carleton University | Sept 2024 – Apr 2025 | Ottawa

- Led office hours to support students struggling with concepts such as recursion, runtime analysis, graph algorithms, combinatorics, dynamic programming, and P versus NP, improving overall academic performance
- Developed supplementary learning material to address areas of struggle and deepen students understanding of complex course content

Software Developer

Data Annotation | Apr 2024 – Aug 2024 | Remote

- Developed software for training LLMs in generating correct and readable code to solve users' problems across multiple programming languages such as Java, Python, SQL, JavaScript, C++ and Bash
- Debugged AI generated code for web development, GUI design, data analysis, and mathematical reasoning to improve model responses for diverse query topics

IT Developer Intern

Transport Canada Executive Office | May 2023 – Apr 2024 | Ottawa

- Led development of a user-friendly agenda planner with automation features that streamlined collaboration and communication among committees
- Provided training and support to end users, achieving seamless adoption of the agenda planner system and positive feedback from committee leads
- Automated the staffing action approval process for the department using Power Automate, reducing approval time by over 75%
- Conducted client requirement gathering sessions to ensure final products included all the desired functionality

Projects

Machine Learning PID Controller – C, ESP-IDF, FreeRTOS, Python

Worked in a team to design and implement a real-time system for PID controller tuning using a genetic algorithm. Developed embedded software on an ESP32 microcontroller with FreeRTOS to control a wheel motor, process encoder input, and dynamically adjust motor behavior through a PID feedback loop. Developed and deployed a genetic algorithm that aimed to optimize PID gain values based on real-world test performance.

Delaunay Triangulation Visualization Tool – Python, Flask, Javascript, HTML/CSS

Implemented the incremental Delaunay Triangulation Algorithm in Python and developed an interactive visualization tool to illustrate the triangles, circumcircles and edge flips of the algorithm. Extended the project into a Flask based web application with a JavaScript frontend for browser-based use. Conducted runtime experiments across multiple input distributions to demonstrate the efficiency of randomization.

Neureset EEG Treatment Simulator – Qt/C++, Git

Collaborated in an Agile team to create a simulation of a medical device which uses brain signal input to diagnose and treat conditions. Led sprint planning meetings, wrote readable and maintainable C++ code, and performed code reviews for seamless integration and consistent quality across the team.

Scheduling Program – CUSERT (Carleton University Student Emergency Response Team) | Java

Created a fully automated scheduling tool using object-oriented programming in Java to track and coordinate each volunteer's schedule. Developed a customized algorithm to meet minimum staffing requirements and reduced the time required for scheduling from several hours to minutes.